

Pandas Data Analysis Documentation

Official Reference Guide to DataFrames, Series, GroupBy, and Transformations

1. DataFrames & Series Operations

Pandas provides data structures (`Series` for 1D and `DataFrame` for 2D labeled data) optimized for data manipulation, cleaning, and analysis using columnar storage and vectorized operations.

```
import pandas as pd

df = pd.DataFrame({
    'department': ['Engineering', 'Engineering', 'Marketing', 'Sales'],
    'employee': ['Alice', 'Bob', 'Charlie', 'David'],
    'salary': [120000, 115000, 95000, 105000]
})

summary = df.groupby('department')['salary'].agg(['mean', 'max', 'count'])
print("Salary Summary:\n", summary)
```

2. Data Filtering & Merging

Pandas enables relational database-style operations including inner/left/right/outer joins using `merge()`, index alignment, and conditional boolean slicing.

```
# Boolean filtering and merging
high_earners = df[df['salary'] >= 110000]
print("High Earners:\n", high_earners[['employee', 'salary']])
```